

Lectures

The Abstraction Stack
Boolean Logic Basics: AND, OR, NOT, XOR Gates
Truth Tables
Boolean Laws & Simplification
Worked Simplification Examples

Motivation for Minimisation (Cost/Power)
Canonical Forms: SOP and POS
Karnaugh Maps
DeMorgan's Theorems and Bubble Pushing
Universal Gates (NAND/NOR) Demonstration
Don't-Cares in K-Maps

From Gates to Reusable Blocks
Multiplexers
Cascading
Demultiplexers
 n -to- 2^n Decoders
Half Adder
Full Adder
Ripple Carry Adder

Number Systems
Binary, Decimal, Hex Conversions
Signed Representations Overview
Two's Complement
Negation and Range
Sign Extension
Binary Addition and Subtraction
Overflow Detection Rules

Combinational vs Sequential Circuits
SR Latch: Structure and Forbidden State
D-Latch (Level-Triggered)
D Flip-Flop (Edge-Triggered)
Setup Time, Hold Time, Metastability

Scaling from Flip-Flops to Registers
Registers: Load Enable, Reset

Asynchronous (Ripple) vs Synchronous Counters
4-bit Synchronous Up-Counter Design
Shift Registers: SISO, SIPO, PISO, PIPO

From Registers to Register Files
Register File Architecture (1 Write, 2 Read)
Introduction to Finite State Machines
Moore vs Mealy Machines
Sequence Detector

SRAM & DRAM

From Gates to a Complete Machine
Von Neumann Architecture: Structure and History
The Von Neumann Bottleneck
Harvard Architecture: Separate I & D Memories
Modified Harvard in Modern CPUs

Labs

Verilog programming of any concept covered in Lectures